

MODEL CONTEXT PROTOCOL · DEEP DIVE

Module 3

Everyday MCP II — Copilot, Other Clients & Popular Servers

VS Code + Copilot agent mode · .vscode/mcp.json · other hosts · popular servers · the MCP Registry

1 HOUR

HANDS-ON LAB

INSTRUCTOR-LED

Workspace vs User Configuration

WORKSPACE (shared with the repo)

- ▶ File: `.vscode/mcp.json` in your project root
- ▶ Scaffold via: **Command Palette → MCP: Open Workspace Folder MCP Configuration**
- ▶ Commit it to git so your team inherits the same MCP servers.

USER (all your workspaces)

- ▶ Open via: **Command Palette → MCP: Open User Configuration**
- ▶ Root key: **also servers (same shape as workspace config)**
- ▶ Use for your personal/private servers — visible across every workspace you open.
- ▶ **Tip:** also installable from the Extensions view — search "@mcp" for the new MCP server gallery.

.vscode/mcp.json — Minimal & Full Shape

```
// Full – stdio + remote + prompted secret
{
  "inputs": [{ "type": "promptString", "id": "gh-pat", "password": true }],
  "servers": {
    "filesystem": {
      "type": "stdio", "command": "npx",
      "args": ["-y", "@modelcontextprotocol/server-filesystem", "${workspaceFolder}"]
    },
    "github": {
      "type": "http",
      "url": "https://api.githubcopilot.com/mcp/"
      "headers": { "Authorization": "Bearer ${input:gh-pat}" }
    }
  }
}
```

Three Things to Remember in VS Code

The three rules — get these right and MCP in VS Code just works

- ▶ **Root key is servers** — never `mcpServers` in `.vscode/mcp.json`. Pasting a Claude config unchanged is the classic load failure.
- ▶ **Pick the Agent in the agent picker** — MCP tools surface in Copilot Chat only when Agent is selected in the agent picker (Ask and Plan agents ignore MCP). Remote MCP + OAuth GA in VS Code 1.101.
- ▶ **Secrets via inputs** — top-level inputs array (type `promptString`, password `true`), reference as `${input:id}`. Never hard-code API keys.
- ▶ **Bonus: `${workspaceFolder}`** — VS Code variable that resolves to your project root; portable across machines and sidesteps Windows backslash issues.

If a server does not start — diagnostics

- ▶ **Output panel → "MCP"** — shows each server's startup logs and stderr; first place to look when the play code-lens reports a failure.
- ▶ **Sanity check in a terminal** — run the exact command (`npx -y @modelcontextprotocol/server-filesystem <dir> · uvx mcp-server-fetch`); if it errors there, fix there first.
 - Developer: Reload Window forces VS Code to re-read `mcp.json`; useful when a stuck state lingers between edits.

Starting Servers & Using Tools in Agent Mode

Three controls drive the daily workflow — the play code-lens, the Tools icon in Copilot Chat, and the Agent-mode dropdown.

Control	What it does	Where it lives
▶ Start code-lens	Inline button above each server entry in mcp.json — click to launch that server	.vscode/mcp.json editor
Tools icon	Shows the count of available tools across all running servers — click to inspect the list	Copilot Chat panel
Configure Tools	Check / uncheck which tools Copilot may use as context this session — least-privilege made easy	Tools popover → Configure

Mode reminder: MCP tools surface only when Agent (or a custom agent) is selected in the agent picker — Ask agents do not expose MCP tools.

Workflow: pick a server entry → click ▶ → switch to Agent → type a natural prompt → approve the tool call.

Hands-On Lab — Two Servers in VS Code Agent Mode

Goal: configure filesystem + fetch in `.vscode/mcp.json`, start them, and drive both from **GitHub Copilot agent mode** — read a local file and fetch a URL. (~25-30 min)

- ▶ **Prereqs:** VS Code 1.102+, GitHub Copilot + Copilot Chat extensions (active subscription), Node.js 18+, uv (uvx)
- ▶ **Recreate the sandbox (Linux/macOS):**

```
mkdir -p "$HOME/mcp-lab-sandbox" && printf 'Hello from MCP – Module 3 VS Code lab.\n' > "$HOME/mcp-lab-sandbox/hello.txt"
```

- ▶ Open the sandbox folder in VS Code → create `.vscode/mcp.json` with the servers root key (MCP: Open Workspace Folder Configuration scaffolds it)
- ▶ Click the ▶ Start code-lens above each server in `mcp.json`; wait for both to show Running
- ▶ Then switch Copilot Chat to **Agent in the agent picker** and drive both — "Read `hello.txt` in this workspace" and "Fetch `modelcontextprotocol.io/llms.txt` and summarize." Approve each tool call.

VS Code Workspace + .vscode/mcp.json

Action Checklist

Three steps to a working .vscode/mcp.json with both reference servers.

- Step 1: File → Open Folder → choose your ~/mcp-lab-sandbox. The folder becomes your VS Code workspace.
- Step 2: Command Palette → MCP: Open Workspace Folder MCP Configuration. VS Code scaffolds .vscode/mcp.json for you (or browse the new gallery: MCP: Browse MCP Servers / @mcp in Extensions view).
- Step 3: Paste the JSON shown on the right. Note the servers root key (NOT mcpServers — that is the #1 mistake).
- \${workspaceFolder} resolves to the opened sandbox — no hard-coded path, no Windows backslash issue.

.vscode/mcp.json

```
// root key is "servers" — NOT mcpServers
{
  "inputs": [],

  "servers": {
    "filesystem": {
      "type": "stdio",

      "command": "npx",
      "args": ["-y",
        "@modelcontextprotocol/server-filesystem",
        "${workspaceFolder}"]
    },

    "fetch": {
      "type": "stdio",
      "command": "uvx",
      "args": ["mcp-server-fetch"]
    }
  }
}
```